29. **Crank Pins:** (a) All new crank pins applied shall have the place and date applied, original diameter and kind of material used, stamped on the end of the pin in accordance with tracing No. D-75490.

(b) An internal inspection of all hollow main crank pins must be made for detail fractures at the monthly and annual inspection periods. When the side rods are removed, an external inspection of all crank pins must be made for detail fractures, giving special attention to the back fillet next to the wheel hub.

(d) Crank pins must be renewed when worn to the following limits:

	Road Limit	Shop Limit
Diameter of rod journal, less than standard	1/4"	7/32"
Rod journal out of round	1/16"	1/32"
Rod journal tapers	1/16"	1/32"
Rod journal longer than standard	3/16"	1/8"
Main rod journal eccentric	1/8"	3/32"

(e) For crank pin mounting pressure see Locomotive Maintenance Instructions No. L-15.

30. **Driving Boxes:** Driving boxes must be maintained in accordance with Locomotive Maintenance Instructions No. L-16. Broken crown bearings must be renewed. Loose crown bearings must be renewed or tightened with not more than one shim.

31. **Driving Box Shoes and Wedges:** When necessary to apply liners to driving box shoes or wedges, a single liner of sufficient thickness, instead of two or more thin liners, must be riveted to each shoe or wedge. The minimum thickness of the wall to which the liner is riveted must be 1/2" after repairs are completed.

32. **Frames, etc.:** Frames, deck plates, tail pieces, pedestals and braces must be cleaned and thoroughly inspected each time the locomotive is in shop for heavy repairs.

33. **Lateral Motion:** (a) Total lateral motion must not exceed the following:

For engine truck wheels (trucks with rigid centers)	1 1/2"
For engine truck wheels (trucks with swing centers)	1"
Trailer Truck wheels	1"
Driving wheels, more than one pair (see note)	3/4"

NOTE—On some D-16 type locomotives the lateral motion will have to be kept below this limit on account of interference with the side sheets.

(b) Lateral motion must be kept within such limits that the driving wheels, rods and crank pins will not interfere with other parts of the locomotive.

34. **Pilots:** The clearance above the rail shall be within the limits of 3 1/2" and 6", as shown on the tracings.

35. **Spring Rigging:** Springs and spring rigging must be maintained approximately level so as to insure the proper distribution of weight to the various wheels of the locomotive. Driving box saddle, equalizer, hanger, bolt or pin, when broken, must be renewed. When reclaimed leaves are used in a spring, the total length must not be more than 3/8" over or under tracing size. Springs with the following defects must be renewed:

(a) Top leaf broken or two leaves in upper half, or any three leaves in spring broken. (The long side of the spring is to be considered the top.)